LAB REPORT

GROUP NUMBER	Group 10
LAB PARTNERS	许维震，胡钦源，郭雨桐 周语涵，黄梓恒，武文翊
LAB DATE	2024.11.23
LAB EXPERIMENT	FTIR and Thermal Analysis Experiments

1.Abstract

In this experiment, we need to identify six substances by using Fourier Transform Infrared Spectroscopy (FTIR) and Thermogravimetric Analysis-Differential Scanning Calorimetry (TG-DSC). By using Fourier Transform Infrared Spectroscopy (FTIR), we can get the functional groups and chemical bonds contained in the substances, so that we can preliminarily determine the types of substances, and then we can further verify them by Thermogravimetric Analysis-Differential Scanning Calorimetry (TG-DSC). Through this experiment, we can get a preliminary understanding of the relationship between the structure and properties of materials, and a preliminary understanding of the methods and principles of the experiment.

2. Introduction

2.1 FTIR (Fourier Transform Infrared Spectroscopy)

FTIR is a technique used to analyze the molecular structure and chemical composition of materials by measuring their absorption of infrared light. When molecules absorb infrared energy, characteristic vibrational transitions occur, which are closely related to the molecular structure. Fourier transform technology converts the interferogram into an infrared spectrum, ultimately presenting the absorption spectrum of the sample. The position, intensity, and shape of characteristic peaks in the spectrum allow the identification of chemical bonds and functional groups.

2.2 TG-DSC (Thermogravimetric Analysis-Differential Scanning Calorimetry)

TG-DSC (Thermogravimetric-Differential Scanning Calorimetry) is a technique that combines Thermogravimetric Analysis (TG) and Differential Scanning Calorimetry (DSC) to study the thermal stability, decomposition processes and phase transition properties of a sample by simultaneously measuring the mass change and thermal effects during heating.Thermogravimetric Analysis (TGA) studies the thermal stability and decomposition behavior of materials by precisely measuring the changes in sample mass as a function of temperature. The sample is exposed to an inert or oxygen atmosphere under controlled heating conditions, and its mass changes reflect processes such as the loss of volatile components, chemical decomposition, and redox reactions. The thermogravimetric curve (mass vs. temperature or time) provides information on decomposition temperatures, reaction rates, and composition.In DSC analysis, on the other hand, the heat absorbed or released by the sample during heating is measured to help identify thermal behaviors such as melting, glass transition, and crystallization.

3. Experimental Objectives

3.1 FTIR

(1). To be able to skillfully use the FTIR experimental equipment

(2). Be able to identify functional groups and chemical bonds by analyzing FTIR images.

3.2 TGA

(1) Knowledge of the use of thermal analyzers

(2) Understand the basic principles of thermogravimetric analysis and differential scanning calorimetry

3.3 Aaterial analysis

There were six materials to be distinguished in the experiment:Citric acid trinitrina dihydrate， Hydroxylamine hydrochloride， Polyvinyl butyral， melamine， thiosemicarbazone， Tris(hydroxymethyl)aminomethane

4.Experimental Steps

4.1FTIR

(1) Sample preparation: The sample is a solid sample, so the sample is simply placed in a mortar and ground to powder form, and then analyzed using the ATR (Attenuated Total Reflectance) accessory.

(2) Instrument Preparation: The FTIR instrument was turned on and warmed up to ensure that the instrument operated in a stable state, and then calibrated with a standard.

(3) Sample Measurement: Place the sample in the sample position of the FTIR instrument, set the appropriate scanning parameters, measure the

infrared spectral data of the sample and record.

(4) Clean up the experimental equipment: After completing the experiment, clean up the instrument and store the sample.

4.2.TG-DSC

(1) Sample preparation: Weigh 0.5mg of sample in the sample disk.

(2) Instrument preparation: Start the instrument and check to ensure that the TG and DSC channels are working properly.

(3) Experimental measurements: Simultaneously place the sample tray with and without sample, in the instrument, to start the experiment, and after a stable run, get the mass change (TG) and temperature-heat flow change (DSC) of the sample

5.Laboratory apparatus

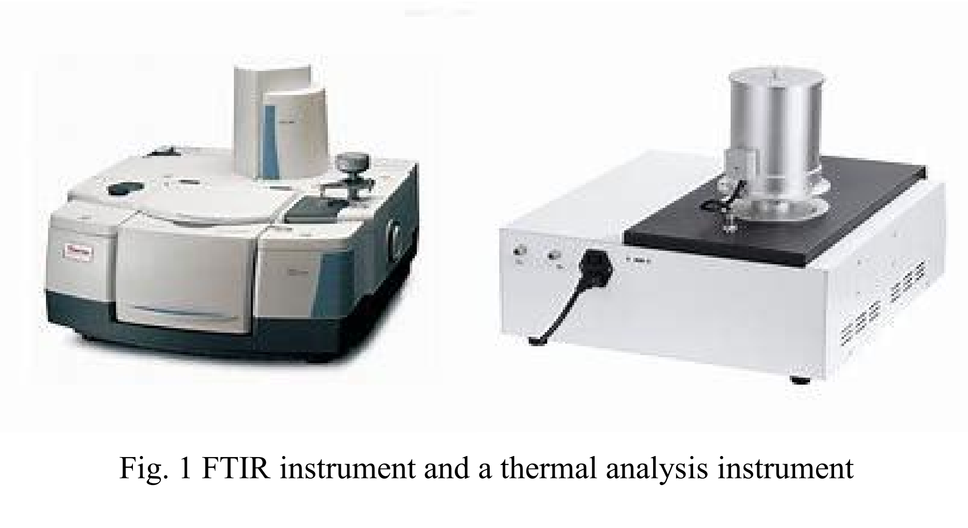

Fig. 1 FTIR instrument and a thermal analysis instrument

6. Analysis of the experimental data

6.1.B1 Experimental data (Qinyuan Hu)

After comparing the infrared spectrum with that of melamine, we can formulate that sample B1 is melamine, and the chemical bonds that melamine has include C=N, C-N, and N-H. The peaks of B1 are summarized as follows.

The positions of the peaks were calibrated according to the absorption frequencies of each chemical bond:

Chemical Bonding and Vibration Types	Intervals in which they are located	Position in the diagram
C=N stretching	1800-1620cm^{-1}	1711cm^{-1}
C-N stretching	1360-1020cm^{-1}	1266cm^{-1}
N-H stretching	3500-3300cm^{-1}	3248cm^{-1}
N-H bending(inside)	1650-1550cm^{-1}	1521cm^{-1}
N-H bending(outside)	900-650cm^{-1}	760cm^{-1}

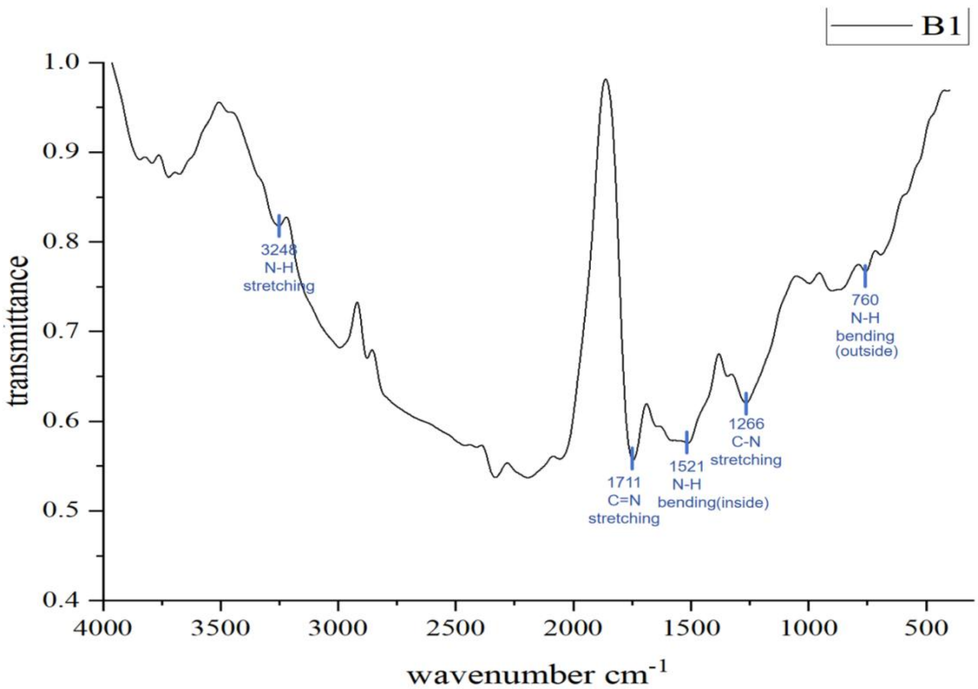

Fig.2 Infrared spectroscopic results of B1

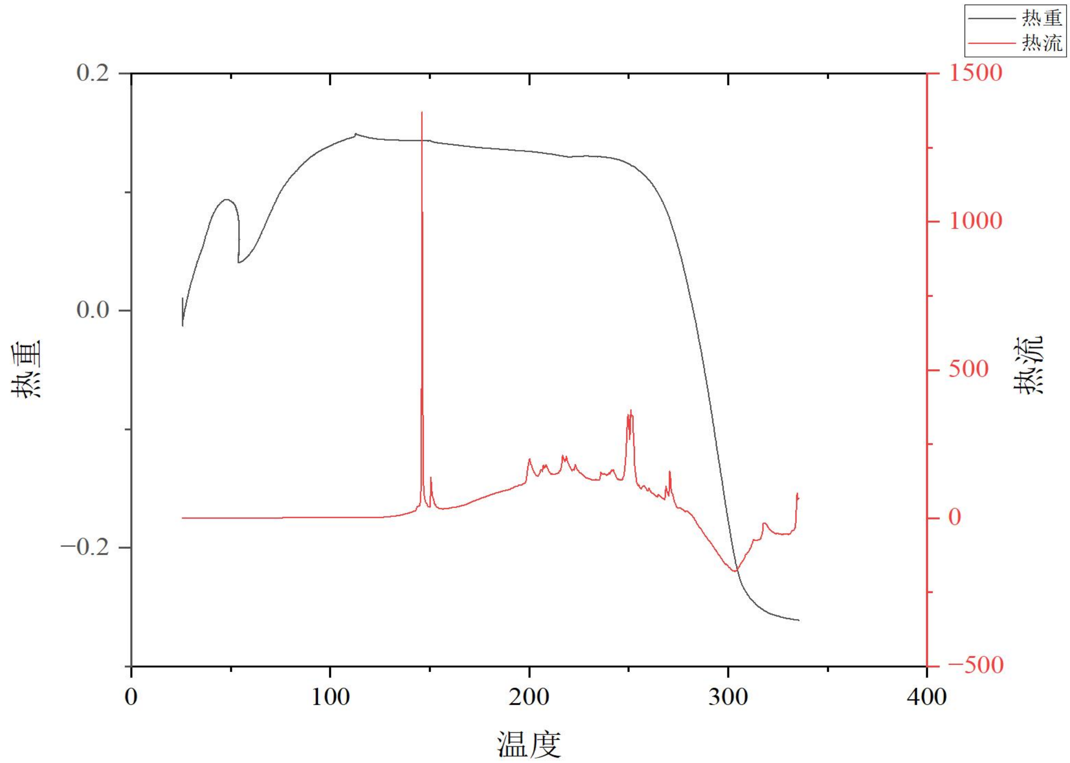

Fig.3 The TG-DSC spectrum of sample B1

TG curve analysis:

0-100°C: small fluctuations in mass, possibly due to evaporation of adsorbed water or adjustment of the initial state of the sample. 100-200°C: the thermogravimetric curve is relatively smooth, indicating that the sample is thermally stable in this temperature range, with no significant decomposition or loss of mass. 200-300°C: a significant decrease in the thermogravimetric curve, indicating that the sample undergoes a thermal decomposition, volatilization, or dehydration, and that the organic components may begin to decompose. Organic components begin to decompose.

DSC curve analysis:

150-200°C: Sharp peaks appear in the heat flow curve, which may be due to noise in the heat flow instrument or a localized exothermic reaction in the sample. 200-300°C: Multiple small peaks appear in the heat flow curve, which suggests that the decomposition reaction of the sample is complex and involves multiple reaction phases. 300°C and onwards: The trend of the heat flow curve is downward, which may be due to the onset of thermal effects accompanying the thermal decomposition. The main exothermic phase is at >250°C: this corresponds to the weight loss in the TG curve, which indicates that energy is released from the decomposition of the sample at this time.

6.2.B2 Experimental data (Weizhen Xu)

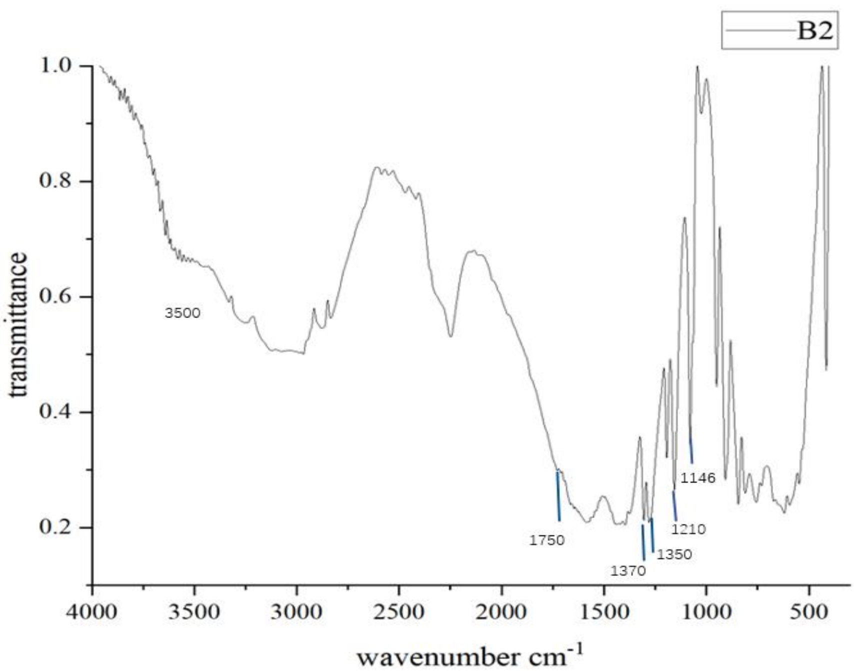

Fig.4 Infrared spectroscopic results of B2

At 1750 cm^{-1} is the stretching vibrational absorption peak of C=O

At 1146 cm^{-1} is the stretching vibrational absorption peak of C-O

At 1350 cm^{-1} is the bending vibrational absorption peak of O-H

At 1210 cm^{-1} is the stretching vibrational absorption peak of C-C

At 1370 cm^{-1} is the bending vibrational absorption peak of C-H

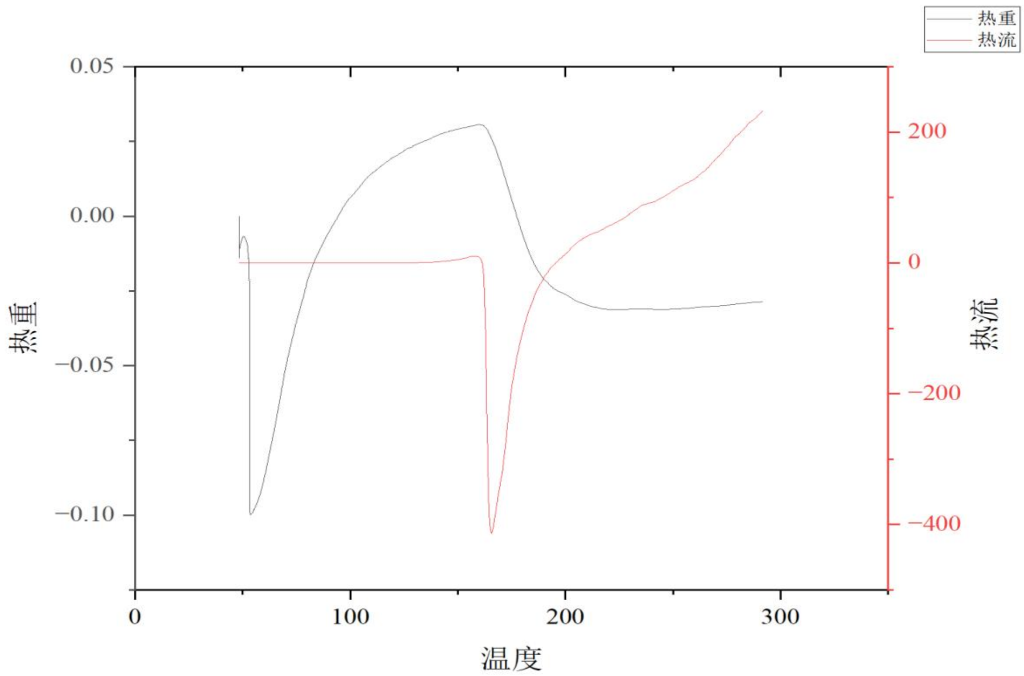

Fig.5 The TG-DSC spectrum of sample B2

According to the provided, thermogravimetric-thermofluorimetric chromatogram, the following detailed analysis follows:

Thermogravimetric Analysis (TGA) Curve Analysis of Thermal Weight Changes:

1. Temperature Range: 0-50°C

The TGA curve shows minor fluctuations, which may be caused by moisture absorption, water evaporation, or adjustments to the initial state of the sample.

2. Temperature Range: 50-150°C

The TGA curve shows a marked decrease, indicating a loss of mass in the sample, which may be due to the removal of water or volatile substances.

3. Temperature Range: 150-250°C

The TGA curve tends to stabilize, with no obvious change in mass of the sample, indicating good thermal stability.

4. Temperature Range: 250-300°C

The TGA curve shows a slight upward trend or fluctuation, which may be due to drift in the experimental instrument or an anomalous reaction process in the sample.

Key Phenomena:

1. Major Mass Loss Stage: Between 50-150°C, the sample shows a marked loss of mass, which is

III. Combining TGA and DSC curves analysis

1. 50-150°C region:

TGA curve: significant weight loss, indicating sample decomposition.

DSC curve: appearance of an endothermic peak, corresponding to sample dehydration or release of volatile components.

2. The temperature range of 150-300°C:

Thermogravimetric curve: tends to be stable, indicating that the sample's mass change is small.

Thermal flow curve: gradually increases, which may indicate a slow thermal decomposition or oxidation reaction of the sample.

3. Overall Phenomenon Summary:

Absorption of Heat and Weight Loss Stage (50-150°C): Removal of water or volatile substances.

Stable Thermal Decomposition Stage (150-300°C): Good thermal stability, no significant mass loss, but there is a minor change in the heat flow curve.

IV. Conclusion

Summary of Sample Behavior:

50-150°C: The heat-absorbing weight loss stage, mainly due to water evaporation or volatile component release.

150-300°C: The sample has good thermal stability, with little change in mass, and the heat flow curve shows a slow thermal effect.

Overall, the spectrum chart shows a clear heat-absorbing weight loss characteristic, with a gradual change in heat flow, indicating good thermal stability of the sample.

6.3.B3 Experimental data(Ziheng Huang)

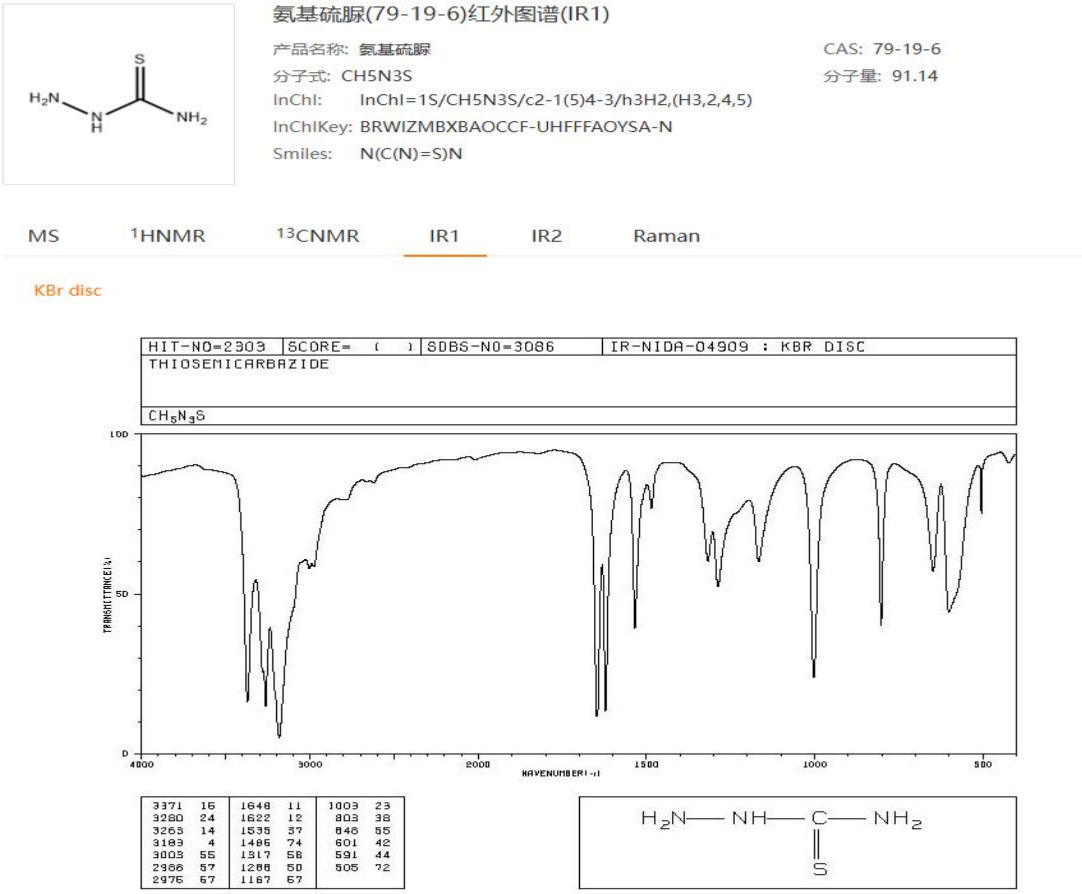

Fig.6.Substance information for Thiosemicarbazide

By analysing the FTIR spectrum of sampleA1 using omnic and comparing the valid peaks, we conclude that the substance in sample B3 is Thiosemicarbazide. Figure 6 shows the material properties of Thiosemicarbazide

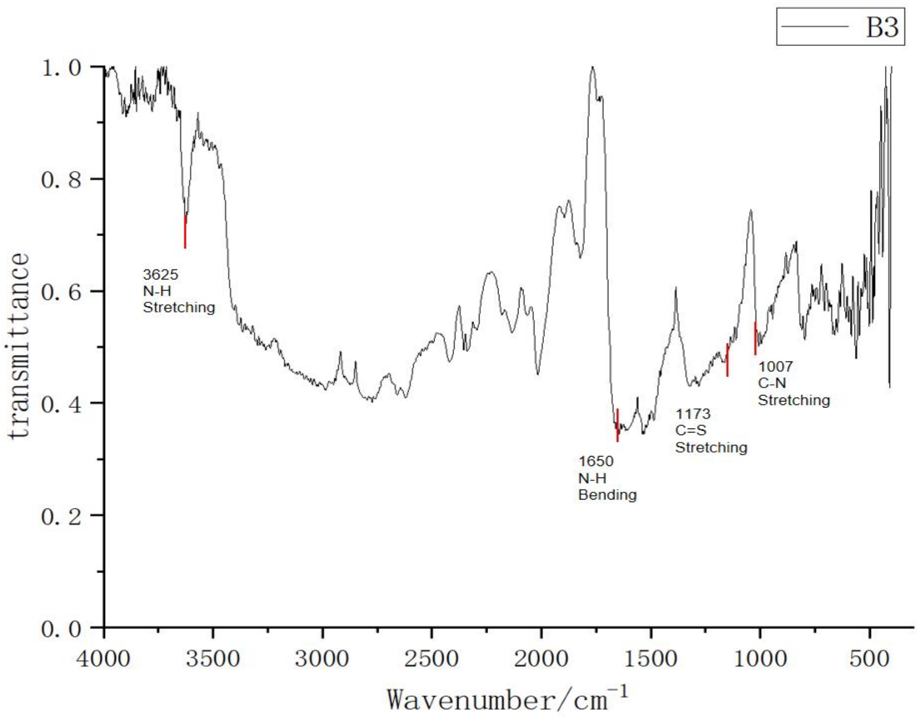

Fig7.The FTIR absorption spectrum of sample B3

Judgement of individual peaks：

At 3625 cm-1 are the absorption peaks of the stretching vibration of N-H

At 1650 cm-1 are the absorption peaks of the bending vibration of N-H

At 1173 cm-1 are the absorption peaks of the stretching vibration of C=S

At 1007 cm-1 are the absorption peaks of the stretching vibration of C-N

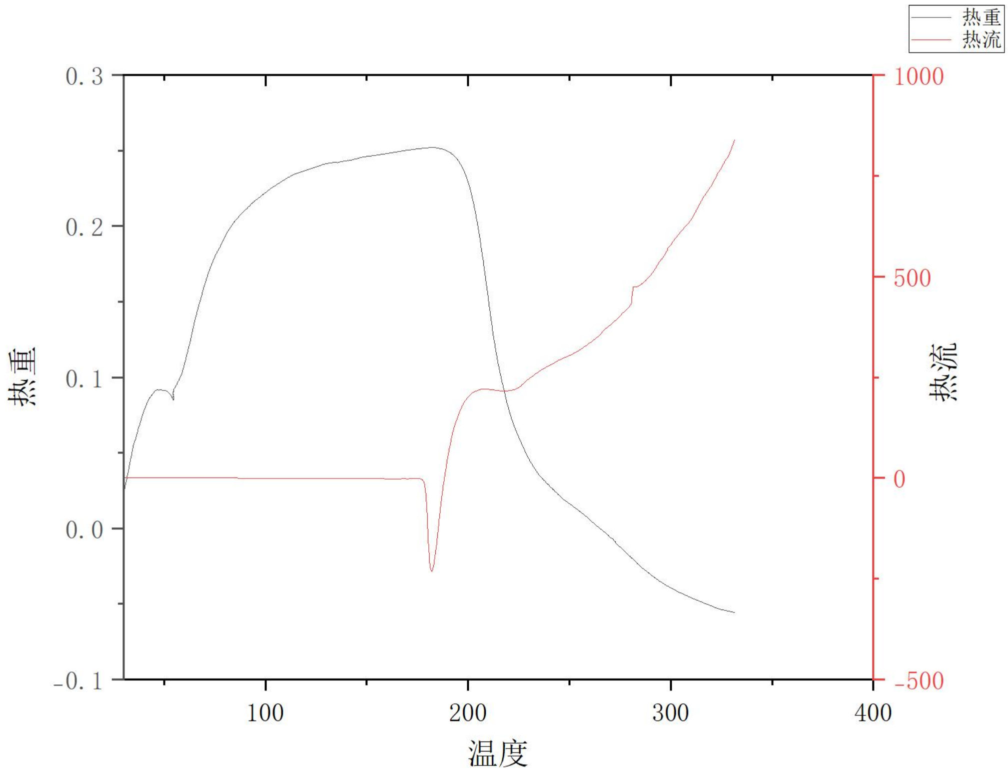

Fig 8. The TG-DSC spectrum of sample B3

Based on the thermogravimetric/heat flow spectrogram, the following analysis can be performed:

I. Thermogravimetric (TGA) curve analysis

1. Vertical coordinate: Thermogravimetric (in mg or normalized scale) on the left side, reflecting the change in sample mass.

2. Thermogravimetric change observation:

At 50-150°C: the thermogravimetric curve rises rapidly, which may indicate absorption of moisture, adsorption, or the effect of instrumental baseline drift (if the mass increases).

From 150-300°C: the thermogravimetric curve shows a slow decrease, which may be due to loss of water, volatilization or partial decomposition of the sample.

After 300°C: the thermogravimetric curve continues to decrease, indicating further decomposition or oxidation of the sample.

3. Critical temperature points:

Mass stabilization stage: approximately 0-50°C and 150-200°C regions with little change in mass.

Significant weight loss phase: mainly in the 200-350°C region, where thermal decomposition or volatilization is presumed to have occurred.

End point residue: after the temperature reaches 400°C, the mass is nearly stabilized, indicating some non-decomposable residue (e.g. ash or inorganic components).

II. Heat flow (DSC) curve analysis

1. vertical coordinate: the right side is heat flow (unit: mW or standardized unit), reflecting the heat absorption or exothermic thermal events.

2. Heat flow peak characteristics:

At 150-200°C: a distinct downward heat absorption peak appears. This indicates that the sample undergoes a **heat absorption reaction**, possibly:

Dehydration of the hydrate (water loss and heat absorption).

Partial thermal decomposition of organic components.

At 200-300°C: a gradual increase in the heat flow signal, probably due to thermal effects during decomposition.

After 300°C: the heat flow curve rises rapidly, showing an exothermic process, which may be an oxidative decomposition of the sample **or a combustion reaction.

3. The location and area of the peak:

Absorption peak (150-200°C): the peak temperature can be quantitatively analyzed for the start and end temperatures of the heat-absorption reaction.

Exothermic trend (>300°C): indicates the release of heat from decomposition or combustion of the sample at high temperatures.

III. Combined thermogravimetric and heat flow curves

1. 200-300°C region:

The thermogravimetric curve shows a loss of mass, indicating that sample decomposition or volatilization has occurred.

At the same time the heat flow curve shows a heat absorption peak, indicating that the process is a heat absorption reaction.

2. After 300°C:

The thermogravimetric curve continues to decrease, indicating continued decomposition of the sample.

The heat flow curve shifts to exothermic, indicating that this stage may be an oxidative decomposition or combustion process. 3.

3. Residue and thermal effects:

The thermogravimetric curve stabilizes after 400°C. The residual mass indicates the non-decomposable components of the sample Heat flow analysis combined with thermogravimetry reveals the correspondence between sample decomposition and exothermic/thermal absorption.

IV. CONCLUSIONS

50-150°C: possible water evaporation or adsorption.

150-200°C: heat absorption process, possibly dehydration or initial decomposition of organic matter.

200-350°C: decomposition of the sample, accompanied by heat absorption and mass loss.

300°C onwards: exothermic reaction, possibly sample oxidation or decomposition and combustion.

6.4.B4 Experimental data(Yutong Guo)

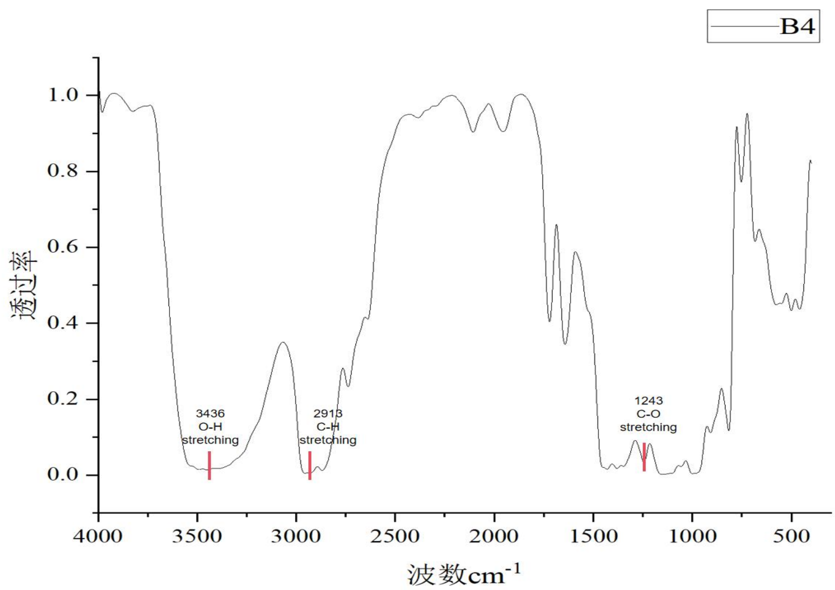

Fig.9 Infrared spectroscopic results of B4

The following functional groups are found by using the charts:

O-H stretching,the standard wave number is 3550-3200cm^{-1},the position in Fig is 3436cm^{-1}.

C-O stretching,the standard wave number is 1270-1010cm^{-1},the position in Fig is 1243cm^{-1}.

C-H stretching,the standard wave number is 3000-2843cm^{-1},the position in Fig is 2913cm^{-1}.

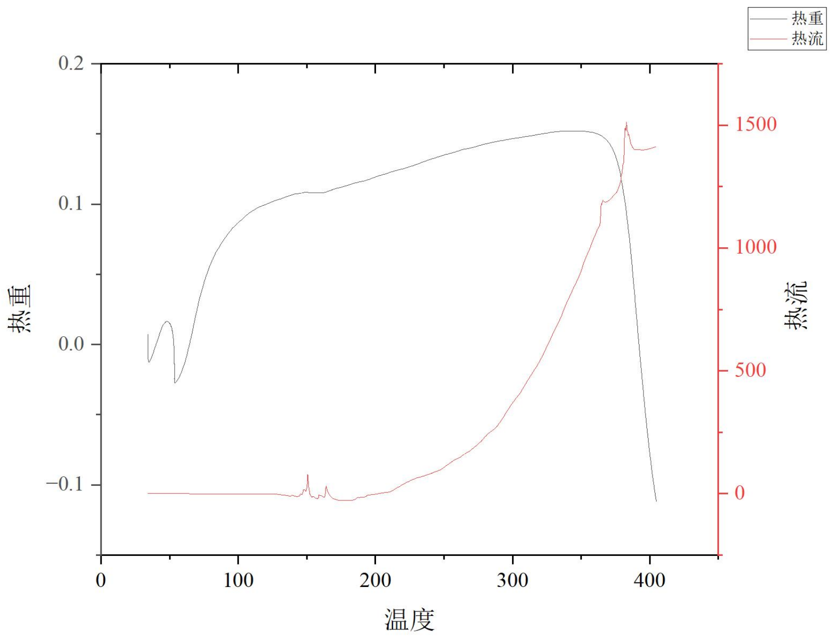

Fig.10 The TG-DSC spectrum of sample B4

The figure shows the trend of thermal weight and heat flow of the material with temperature. As the temperature rises:

I. Thermal Weight (TG) Changes

1.Initial stage: Starting at 0°C, the thermal weight remains low, indicating minimal heat absorption.

2.Temperature Rise Stage: As temperature increases, thermal weight gradually rises. At 100°C, it's near 0.0; at 200°C, it reaches 0.1; at 300°C and 400°C, it continues to increase to 0.2 and beyond.

3.Peak Phase: At a specific temperature, thermal weight peaks, suggesting a thermochemical reaction and rapid heat absorption.

4.Depreciation Stage: Post-peak, with further temperature increase, thermal weight decreases, signaling weakened heat absorption.

II. Heat Flow (DTG) Changes

1.Initial stage: Heat flow remains low as temperature rises from 0°C.

2.Temperature Rise Stage: As temperature increases, heat flow rises gradually. From 0°C to 100°C, it increases to a positive value, and continues to rise at a faster rate at higher temperatures.

3.Peak Stage: Heat flow reaches a peak at a specific temperature, reflecting the maximum heat transfer rate.

4.Decline Stage: After the peak, with continued temperature rise, heat flow decreases, indicating a reduction in heat transfer rate

6.5.B5 Experimental data(Yuhan Zhou)

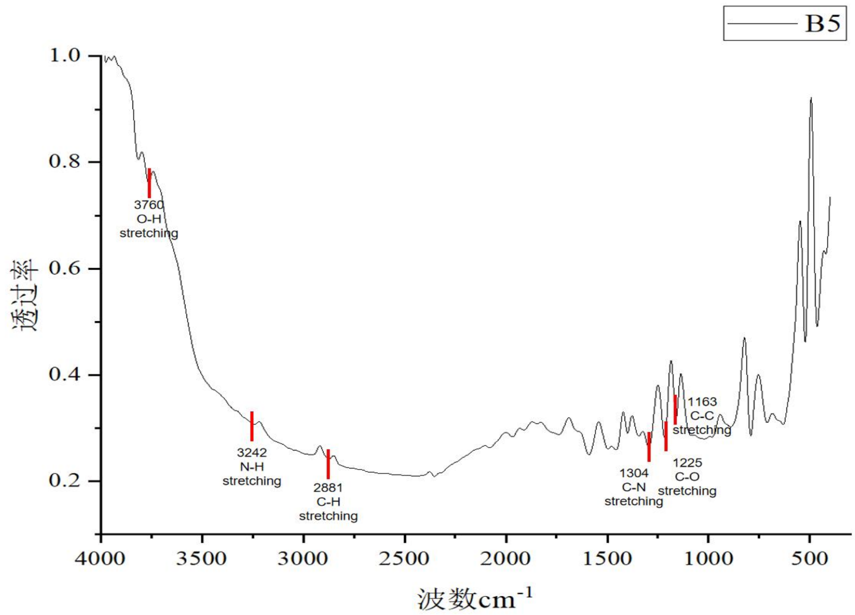

Fig.11 Infrared spectroscopic results of B5

The following functional groups are found by using the charts:

P-H stretching,the standard wave number is 3700-3200cm^{-1}**,**the position in Fig is 3760cm^{-1}**.**

N-H stretching,the standard wave number is 3500-3300cm^{-1}**,**the position in Fig is 3242cm^{-1}**.**

C-H stretching,the standard wave number is 3000-2843cm^{-1}**,**the position in Fig is 2881cm^{-1}**.**

C-N stretching,the standard wave number is 1340-1020cm^{-1}**,**the position in Fig is 1304cm^{-1}**.**

C-O stretching,the standard wave number is 1260-1000cm^{-1}**,**the position in Fig is 1225cm^{-1}**.**

C-C stretching,the standard wave number is 1250-1140cm^{-1}**,**the position in Fig is 1163cm^{-1}**.**

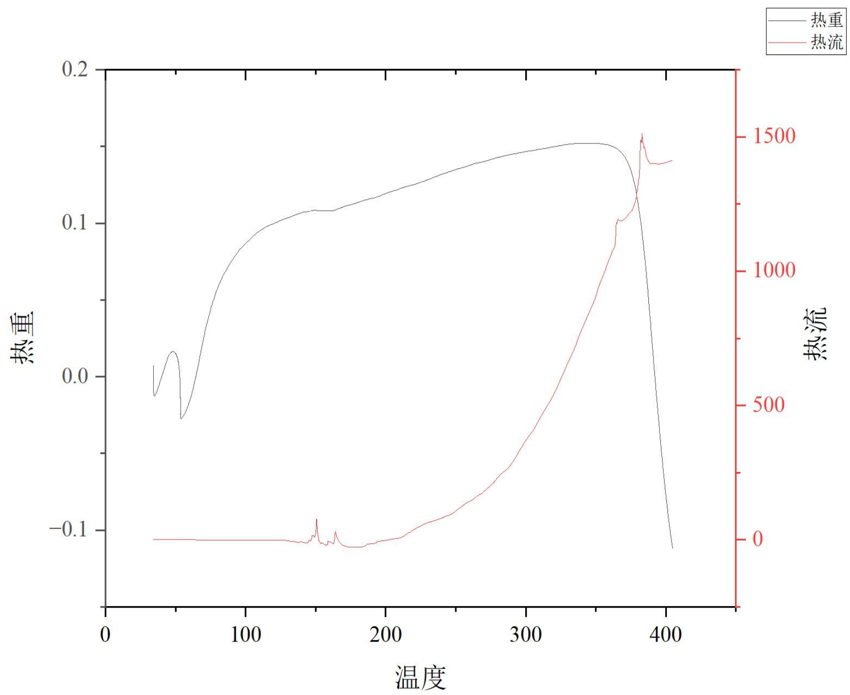

Fig.12 The TG-DSC spectrum of sample B5

The figure shows the trend of thermal weight and heat flow of the material with temperature. As the temperature rises:

I. Thermal Weight (TG) Changes

1.Initial stage: Starting at 0°C, the thermal weight remains low, indicating minimal heat absorption.

2.Temperature Rise Stage: As temperature increases, thermal weight gradually rises. At 100°C, it's near 0.0; at 200°C, it reaches 0.1; at 300°C and 400°C, it continues to increase to 0.2 and beyond.

3.Peak Phase: At a specific temperature, thermal weight peaks, suggesting a thermochemical reaction and rapid heat absorption.

4.Depreciation Stage: Post-peak, with further temperature increase, thermal weight decreases, signaling weakened heat absorption.

II. Heat Flow (DTG) Changes

1.Initial stage: Heat flow remains low as temperature rises from 0°C.

2.Temperature Rise Stage: As temperature increases, heat flow rises gradually. From 0°C to 100°C, it increases to a positive value, and continues to rise at a faster rate at higher temperatures.

3.Peak Stage: Heat flow reaches a peak at a specific temperature, reflecting the maximum heat transfer rate.

4.Decline Stage: After the peak, with continued temperature rise, heat flow decreases, indicating a reduction in heat transfer rate.

6.6.B6 Experimental data(Wenyi Wu)

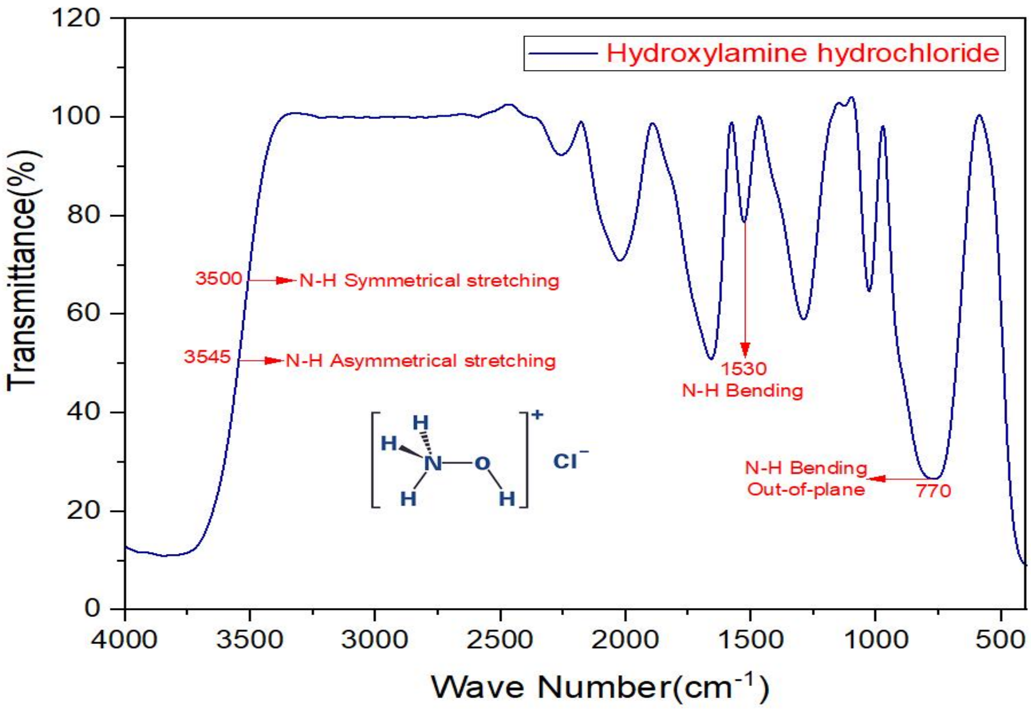

Fig.13 Infrared spectroscopic results of B6

The FTIR spectrum of the sample is shown in the above figure. 3545 cm^{-1} is the asymmetrical stretching vibration of N-H, and 3500cm^{-1} is the symmetrical stretching vibration of N-H. 1530cm^{-1} is the bending vibration of N-H. 770 cm^{-1} is the out-of-plane bending vibration of N-H. In summary, it is easy to obtain the presence of NH^3 .After the comparative analysis of the spectrum, it was concluded that the substance should be Hydroxylamine hydrochloride.

Although the heat heat flow data are not provided, the material name of B6 can be analyzed by the experimental results of the remaining personnel in the group.

7. Conclusion

After the above analysis of the FTIR and TG-DSC of each substance, we can finally determine their names.

Order number	The name of the material
B1	melamine
B2	Citric acid trinitrina dihydrate
B3	thiosemicarbazone
B4	Polyvinyl butyral
B5	Tris(hydroxymethyl)aminomethane
B6	Hydroxylamine hydrochloride

During the experiment, we fully understand the application of Fourier transform infrared spectroscopy and synchronous thermal analysis in materials science, and understand the basic data processing methods. In conclusion, we think we have basically achieved the goals of this experiment.

8. Division of personnel

Group members	Sample analysis
Weizhen Xu	**Analyse B2**
Qinyuan Hu	**Analyse B1**
Yutong Guo	**Analyse B4**
Yuhan Zhou	**Analyse B5**
Ziheng Huang	**Analyse B3**
Wenyi Wu	**Analyse B6**

The rest of the work was completed together by the group.